3. Pressurized or Pump circulation cooling system

### 7.2.1 Direct (or) Non-Return System

This method is mostly used where availability of water is more such as in big industrial engines, marine engines etc., In this method the water from the storage tank is sent to the water jackets of the engine where the water absorbs the heat and the hot water is sent out through the outlet path. The pump in the marine engines pumps the sea water into the water jacket and sends the hot water out. This type is not recommended for automotive engines.

### 7.2.2 Thermosyphon System

We know that the density of hot water is less than that of the cold water. Thermosyphon cooling system works on the principle of circulating water by the density difference in the water. Figure 7.2 shows the thermosyphon system.

In Thermosyphon cooling system, when the engine runs, the hot water in the engine water jacket moves upwards due to the reduction in density. It moves further to the radiator through the hose provided and it is cooled in the radiator. The cold water at the bottom of the radiator enters into the engine and pushes the hot water upwards and occupies the place of the hot

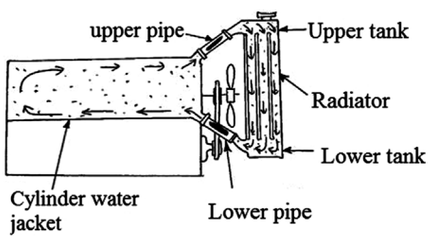



**Figure 7.2.2 Thermosyphon system**

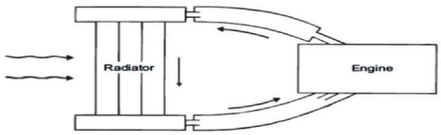



**Figure 7.2.2 Thermosyphon system**

water. This is called as conventional flow of water. In this type of cooling systems the radiator is placed just above the engine. When the water level is less and when the vehicle claims up or down this system does not work properly. Hence this type of cooling system is not is not used in modern automobiles.

### 7.2.3 Pump Circulation System

In this system a water pump is used to circulate the water. Hence this method is also called as pump circulation system. Nowadays, in many vehicles this type of cooling system is followed. In this system the conventional flow of water and the pressure of the centrifugal pump are combined and hence the system has the added advantage. Fig 7.2.3 shows the pump circulation cooling system.

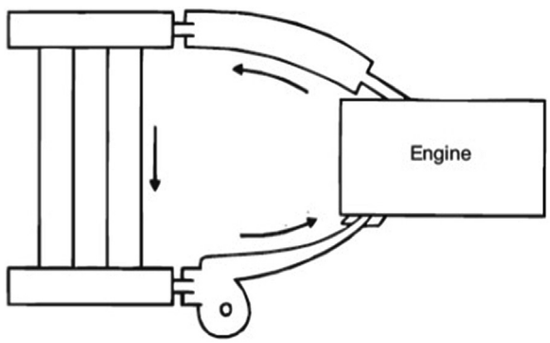



**Fig 7.2.3 pump circulation system**



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